

A Spectral-Norm Improvement for Multiband Approximation of Fourier-Sparse Signals

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Abstract

We revisit a clustering argument used to approximate a k -Fourier-sparse signal by a multiband signal. The usual proof converts pairwise almost-orthogonality into simultaneous energy additivity through an absolute row-sum estimate. We replace that step by a dyadic block bound on the spectral norm of the full correlation envelope. Under the same localizer hypotheses and the same minimum-size clustering rule, this reduces the required separation scale to

$$O\left(\frac{k \log(k/\epsilon) \sqrt{\log k}}{\epsilon}\right)$$

and yields a multiband cover of total width

$$O\left(\frac{k^2 \log(k/\epsilon) (\log k)^{3/2}}{\epsilon}\right).$$

We also give a matching lower example for the abstract correlation envelope and explain why a pure product-size merge threshold does not improve this framework.

1 Overview

Let

$$x(t) = \sum_{j=1}^k \alpha_j e^{2\pi i f_j t}$$

be observed on $[-1, 1]$. As in the source manuscript, write

$$M := M_{k,\epsilon}, \quad \Delta_k := \Delta_{k,\epsilon}.$$

The compact-frequency localizer M converts x into an $L_2(\mathbb{R})$ signal while approximately preserving its energy on $[-1, 1]$. Its Fourier support around each frequency has radius

$$\Delta_k = O\left(\frac{k^2}{\epsilon} \log \frac{k}{\epsilon}\right).$$

Applying this global localizer separately around all k frequencies gives total support width $\tilde{O}(k^3/\epsilon)$. Clustering nearby frequencies and replacing the global localizer by a smaller localizer on each cluster reduces the width to $\tilde{O}(k^2/\epsilon)$.

The central issue is simultaneous almost-orthogonality. Pairwise estimates give a correlation envelope proportional to

$$\frac{|\mathcal{C}_i||\mathcal{C}_j|}{\text{dist}(\mathcal{C}_i, \mathcal{C}_j)}.$$

An absolute row-sum estimate loses two combinatorial logarithms. Our main technical lemma bounds the operator norm of this envelope directly and loses only $\sqrt{\log k}$. The remainder of the multiband argument is unchanged.

All constants below are absolute and may depend on the constant in the localizer interface. As in the source manuscript, we work in the regime $k \gg 1$, use $\log u := \log_2 u$, and suppress constant-factor retuning of ϵ . Thus an intermediate $O(\epsilon)$ error is converted into an ϵ error by running the filters with a sufficiently small constant multiple of ϵ .

2 Setting and localizer interface

We combine repeated frequencies in advance, so the f_j 's are distinct. For a cluster $\mathcal{C}$ drawn from the k frequency-coefficient pairs, define

$$x_{\mathcal{C}}(t) := \sum_{(f, \alpha) \in \mathcal{C}} \alpha e^{2\pi i f t}, \quad |\mathcal{C}| := \#\mathcal{C},$$

and

$$\text{range}(\mathcal{C}) := \left[\min_{(f, \alpha) \in \mathcal{C}} f, \max_{(f, \alpha) \in \mathcal{C}} f \right].$$

When two cluster ranges are disjoint, $\text{dist}(\mathcal{C}, \mathcal{D})$ denotes the distance between them. We write $\|u\|_{[-1, 1]}$ for the L_2 -norm on $[-1, 1]$, following the source manuscript.

We isolate the analytic input needed by the clustering argument.

Definition 2.1 (Admissible localizers). *Fix an absolute constant $A \geq 1$. A family $\{M_{\ell, \delta} : \ell \geq 1, 0 < \delta \leq 1/2\}$ is admissible if the following properties hold.*

(L1) *For every ℓ -Fourier-sparse signal u ,*

$$\|M_{\ell, \delta} \cdot u - u\|_{[-1, 1]}^2 \leq A\delta \|u\|_{[-1, 1]}^2, \quad \|M_{\ell, \delta} \cdot u\|_{\mathbb{R} \setminus [-1, 1]}^2 \leq A\delta \|u\|_{[-1, 1]}^2.$$

(L2) *The Fourier transform of $M_{\ell, \delta}$ is supported in $[-\Delta_{\ell, \delta}, \Delta_{\ell, \delta}]$, where*

$$\Delta_{\ell, \delta} \leq A \frac{\ell^2}{\delta} \log \frac{\ell}{\delta}.$$

Consequently, if the frequencies of u lie in an interval J , then $\text{supp}(\widehat{M_{\ell, \delta} \cdot u}) \subseteq J \pm \Delta_{\ell, \delta}$.

(L3) *For $\ell \leq k$, $0 < \epsilon \leq 1/2$, and $\delta = \epsilon\ell/k$, every ℓ -Fourier-sparse signal u satisfies*

$$\|M_{\ell, \delta} \cdot u - M \cdot u\|_2^2 \leq A\delta \|M \cdot u\|_2^2.$$

(L4) *Let u and v have Fourier sparsities at most a and b , respectively, where $a, b \leq k$, and let their frequency separation be d . For $0 < \delta \leq 1/2$, if*

$$d \geq A \frac{ab}{\delta} \left(\log \frac{ab}{\delta} + \log \frac{k}{\epsilon} \right),$$

then

$$|\langle M \cdot u, M \cdot v \rangle| \leq \delta \|M \cdot u\|_2 \|M \cdot v\|_2.$$

If $d \geq 2\Delta_k$, the same inner product is zero.

The localizer construction in the source manuscript is intended to establish Theorem 2.1. The contribution of this note begins with the clustering and linear-algebraic consequences of that interface.

3 Clustering and its range cost

For a separation parameter $d_{\min} > 0$, run the following analysis-only clustering procedure.

1. Begin with one singleton cluster for each frequency, ordered from left to right.
2. While there are clusters $\mathcal{C}$ to the left of $\mathcal{D}$ such that

$$\text{dist}(\mathcal{C}, \mathcal{D}) \leq \min\{d_{\min} \min\{|\mathcal{C}|, |\mathcal{D}|\}, 2\Delta_k\}, \quad (3.1)$$

replace $\mathcal{C}$, $\mathcal{D}$, and every cluster between them by their union.

3. Return the remaining ordered clusters.

The procedure terminates because every merge decreases the number of clusters. Its output is a partition into contiguous frequency blocks and satisfies the reverse strict inequality in (3.1) for every pair of distinct output clusters.

Lemma 3.1 (Range created by clustering). *Let $\mathcal{C}_1, \dots, \mathcal{C}_n$ be the output clusters. Then*

$$\sum_{i=1}^n |\text{range}(\mathcal{C}_i)| \leq C d_{\min} k \log k. \quad (3.2)$$

Proof. For positive integers a, b , define

$$c(a, b) := \min\{d_{\min} \min(a, b), 2\Delta_k\}.$$

Define $R_1^* := 0$, and, for every $m \geq 2$, define

$$R_m^* := \max_{1 \leq a < m} \{R_a^* + R_{m-a}^* + c(a, m-a)\}. \quad (3.3)$$

We prove first that R_m^* bounds the range of every size- m cluster created by the algorithm.

The sequence $(R_m^*)_{m \geq 1}$ is nondecreasing. Indeed, choosing $a = m$ in the recurrence for R_{m+1}^* gives

$$R_{m+1}^* \geq R_m^* + R_1^* + c(m, 1) \geq R_m^*. \quad (3.4)$$

We now argue by strong induction on m . The assertion is immediate for a singleton. Suppose a merge is triggered by a left endpoint cluster $\mathcal{C}^L$ and a right endpoint cluster $\mathcal{C}^R$, of sizes a and b , respectively. Let $\mathcal{C}$ be the union of these endpoint clusters and all clusters between them, and put $m := |\mathcal{C}|$. Then $m \geq a + b$. Because $\mathcal{C}^L$ and $\mathcal{C}^R$ contain the leftmost and rightmost frequencies of $\mathcal{C}$, respectively,

$$|\text{range}(\mathcal{C})| = |\text{range}(\mathcal{C}^L)| + \text{dist}(\mathcal{C}^L, \mathcal{C}^R) + |\text{range}(\mathcal{C}^R)|.$$

The merge condition, the induction hypothesis, and (3.4) therefore imply

$$\begin{aligned} |\text{range}(\mathcal{C})| &\leq R_a^* + R_b^* + c(a, b) \\ &\leq R_a^* + R_{m-a}^* + c(a, m-a) \\ &\leq R_m^*. \end{aligned} \tag{3.5}$$

Here the second inequality uses $b \leq m-a$, the monotonicity of R^* , and the monotonicity of $c(a, \cdot)$. This proves the claimed domination of the multiway algorithm by (3.3).

It remains to bound R_m^* . A term in (3.3) is represented by a rooted binary tree with m leaves. An internal node whose two child subtrees have $a \leq b$ leaves contributes $c(a, b)$. Charge this amount uniformly to the a leaves in the smaller child. The charge received by each such leaf is

$$\frac{c(a, b)}{a} = \min \left\{ d_{\min}, \frac{2\Delta_k}{a} \right\} \leq d_{\min}. \tag{3.6}$$

Fix a leaf, and let $a_1, a_2, \dots, a_t$ be the sizes of the smaller-child subtrees on the internal nodes at which that leaf is charged, listed from the leaf toward the root. After the charge corresponding to a_j , the parent subtree has at least $2a_j$ leaves. The child subtree containing the fixed leaf at its next charged ancestor contains this parent subtree. Consequently,

$$a_{j+1} \geq 2a_j \quad (1 \leq j < t).$$

Since $a_t \leq m$, we have $t \leq 1 + \lfloor \log m \rfloor$. Thus each leaf receives total charge at most $d_{\min}(1 + \lfloor \log m \rfloor)$, and summing over all m leaves gives

$$R_m^* \leq d_{\min} m (1 + \lfloor \log m \rfloor) \leq 2d_{\min} m \log m \quad (m \geq 2). \tag{3.7}$$

Finally, put $m_i := |\mathcal{C}_i|$ for the output clusters. Singleton clusters have zero range, and $\sum_i m_i = k$. Therefore

$$\sum_{i=1}^n |\text{range}(\mathcal{C}_i)| \leq 2d_{\min} \sum_{i:m_i \geq 2} m_i \log m_i \leq 2d_{\min} k \log k,$$

which proves (3.2). $\square$

4 The correlation envelope

For the new matrix argument only, abbreviate $s_i := |\mathcal{C}_i|$ and $d_{i,j} := \text{dist}(\mathcal{C}_i, \mathcal{C}_j)$. Define the symmetric matrix

$$K_{ii} := 0, \quad K_{ij} := \begin{cases} \frac{s_i s_j}{d_{i,j}}, & d_{i,j} < 2\Delta_k, \\ 0, & d_{i,j} \geq 2\Delta_k. \end{cases} \tag{4.1}$$

Only pairs with overlapping globally filtered Fourier supports appear in this matrix.

For a vector $v \in \mathbb{C}^n$, let $\|v\|_2 = (\sum_i |v_i|^2)^{1/2}$. For a matrix A , its induced Euclidean operator norm is

$$\|A\|_{2 \rightarrow 2} := \sup_{v \neq 0} \frac{\|Av\|_2}{\|v\|_2} = \sup_{\|v\|_2=1} \|Av\|_2. \tag{4.2}$$

Equivalently, $\|A\|_{2 \rightarrow 2}$ is the largest singular value of A ; if A is symmetric, it equals the largest absolute value of an eigenvalue. We also use the Frobenius norm

$$\|A\|_F := \left(\sum_{i,j} |A_{ij}|^2 \right)^{1/2}, \quad \|A\|_{2 \rightarrow 2} \leq \|A\|_F.$$

Lemma 4.1 (Spectral norm of the envelope). *Suppose $\sum_i s_i \leq k$, the cluster ranges are disjoint and ordered, and every distinct pair with $d_{i,j} < 2\Delta_k$ satisfies*

$$d_{i,j} \geq d_{\min} \min\{s_i, s_j\}. \quad (4.3)$$

Then

$$\|K\|_{2 \rightarrow 2} \leq C \frac{k\sqrt{\log k}}{d_{\min}}. \quad (4.4)$$

The same estimate holds for every principal submatrix of K .

Proof. Put $q_{\max} := \lceil \log k \rceil$, and partition the indices into dyadic size classes

$$\mathcal{I}_q := \{i : 2^q \leq s_i < 2^{q+1}\}, \quad 0 \leq q \leq q_{\max}.$$

For $0 \leq q, r \leq q_{\max}$, write

$$K^{(q,r)} := K[\mathcal{I}_q, \mathcal{I}_r]$$

for the corresponding rectangular block of K .

Packing within one pair of size classes. Fix $q \leq r$, put $s := 2^q$ and $u := 2^r$, and fix $j \in \mathcal{I}_r$. We first prove

$$\sum_{\substack{i \in \mathcal{I}_q \\ i \neq j, d_{i,j} < 2\Delta_k}} \frac{1}{d_{i,j}^2} \leq \frac{\pi^2}{3d_{\min}^2 s^2}. \quad (4.5)$$

Consider first the indices on the right of $\mathcal{C}_j$, ordered by increasing distance from $\mathcal{C}_j$, and denote them by $i_1, \dots, i_p$. By (4.3),

$$d_{i_1,j} \geq d_{\min} s. \quad (4.6)$$

For $1 \leq h < p$, the ranges occur in the order $\mathcal{C}_j, \mathcal{C}_{i_h}, \mathcal{C}_{i_{h+1}}$. Moreover,

$$d_{i_h, i_{h+1}} \leq d_{i_{h+1}, j} < 2\Delta_k.$$

Hence (4.3) applies to $\mathcal{C}_{i_h}$ and $\mathcal{C}_{i_{h+1}}$, and gives

$$d_{i_h, i_{h+1}} \geq d_{\min} s.$$

The ordering of the ranges then implies

$$d_{i_{h+1}, j} \geq d_{i_h, j} + d_{i_h, i_{h+1}} \geq d_{i_h, j} + d_{\min} s.$$

Together with (4.6), this yields $d_{i_h, j} \geq h d_{\min} s$. Therefore the contribution from the right of $\mathcal{C}_j$ is at most

$$\frac{1}{d_{\min}^2 s^2} \sum_{h=1}^{\infty} \frac{1}{h^2} = \frac{\pi^2}{6d_{\min}^2 s^2}.$$

The same argument applies on the left, proving (4.5).

Since $s_i < 2s$ for $i \in \mathcal{I}_q$ and $s_j < 2u$, (4.5) implies

$$\sum_{i \in \mathcal{I}_q} |K_{ij}|^2 = s_j^2 \sum_{\substack{i \in \mathcal{I}_q \\ i \neq j, d_{i,j} < 2\Delta_k}} \frac{s_i^2}{d_{i,j}^2} \leq C \frac{u^2}{d_{\min}^2}. \quad (4.7)$$

Furthermore, $|\mathcal{I}_r|u \leq \sum_{j \in \mathcal{I}_r} s_j \leq k$. Summing (4.7) over $j \in \mathcal{I}_r$ gives

$$\left\| K^{(q,r)} \right\|_{2 \rightarrow 2}^2 \leq \left\| K^{(q,r)} \right\|_F^2 \leq C \frac{ku}{d_{\min}^2}. \quad (4.8)$$

If $q > r$, apply the preceding estimate to the transpose block $K^{(r,q)} = (K^{(q,r)})^\top$. Consequently, for all q, r ,

$$\left\| K^{(q,r)} \right\|_{2 \rightarrow 2} \leq B_{qr}, \quad B_{qr} := C \frac{\sqrt{k} 2^{\max(q,r)}}{d_{\min}}. \quad (4.9)$$

Aggregation of the blocks. For $v \in \mathbb{C}^n$, let $v^{(r)}$ be its restriction to $\mathcal{I}_r$, and put $w_r := \|v^{(r)}\|_2$. By the triangle inequality and (4.9), for each q ,

$$\begin{aligned} \left\| (Kv)^{(q)} \right\|_2 &= \left\| \sum_{r=0}^{q_{\max}} K^{(q,r)} v^{(r)} \right\|_2 \\ &\leq \sum_{r=0}^{q_{\max}} \left\| K^{(q,r)} \right\|_{2 \rightarrow 2} \left\| v^{(r)} \right\|_2 \\ &\leq \sum_{r=0}^{q_{\max}} B_{qr} w_r. \end{aligned}$$

It follows that

$$\|Kv\|_2^2 \leq \|Bw\|_2^2 \leq \|B\|_{2 \rightarrow 2}^2 \|w\|_2^2 = \|B\|_{2 \rightarrow 2}^2 \|v\|_2^2. \quad (4.10)$$

Taking the supremum over $v \neq 0$ gives $\|K\|_{2 \rightarrow 2} \leq \|B\|_{2 \rightarrow 2}$. Finally,

$$\begin{aligned} \|B\|_{2 \rightarrow 2}^2 &\leq \|B\|_F^2 \\ &\leq \frac{Ck}{d_{\min}^2} \sum_{q,r=0}^{q_{\max}} 2^{\max(q,r)} \\ &= \frac{Ck}{d_{\min}^2} \sum_{m=0}^{q_{\max}} (2m+1) 2^m \\ &\leq \frac{Ck}{d_{\min}^2} q_{\max} 2^{q_{\max}} \leq C \frac{k^2 \log k}{d_{\min}^2}. \end{aligned} \quad (4.11)$$

Combining (4.10) and (4.11) proves (4.4).

If $S \subseteq [n]$, repeat the same argument with the classes $\mathcal{I}_q \cap S$. Both the packing estimate (4.5) and the mass bound $\sum_{i \in S} s_i \leq k$ remain valid. Hence the same estimate holds for the principal submatrix K_S . $\square$

5 Simultaneous energy additivity

The next lemma converts the analytic pairwise estimate into an entrywise multiple of K .

Lemma 5.1 (Pairwise envelope). *Let the localizers be admissible, and suppose*

$$d_{\min} \geq C \frac{k \log(k/\epsilon)}{\epsilon}. \quad (5.1)$$

For distinct output clusters, write $y_i := M \cdot x_{\mathcal{C}_i}$. Then

$$|\langle y_i, y_j \rangle| \leq C \log(k/\epsilon) K_{ij} \|y_i\|_2 \|y_j\|_2. \quad (5.2)$$

Proof. The claim is immediate when $d_{i,j} \geq 2\Delta_k$. Otherwise set

$$\delta_{i,j} := C_1 \log(k/\epsilon) \frac{s_i s_j}{d_{i,j}}.$$

The terminal separation and (5.1) imply $\delta_{i,j} \leq 1/2$ when the constant in (5.1) is large enough. Moreover,

$$\frac{s_i s_j}{\delta_{i,j}} = \frac{d_{i,j}}{C_1 \log(k/\epsilon)}.$$

Since $d_{i,j} < 2\Delta_k$ and $\Delta_k = O(k^2 \log(k/\epsilon)/\epsilon)$,

$$\log \frac{s_i s_j}{\delta_{i,j}} + \log \frac{k}{\epsilon} \leq C \log(k/\epsilon).$$

Choosing C_1 sufficiently large makes the separation condition in Item (L4) valid with $\delta = \delta_{i,j}$, proving (5.2). $\square$

Theorem 5.2 (Uniform cluster-energy decomposition). *Choose*

$$d_{\min} := \frac{C_H k \log(k/\epsilon) \sqrt{\log k}}{\epsilon} \quad (5.3)$$

for a sufficiently large absolute constant C_H . Then every subset $S \subseteq [n]$ satisfies

$$\left\| M \cdot \sum_{i \in S} x_{\mathcal{C}_i} \right\|_2^2 = (1 \pm O(\epsilon)) \sum_{i \in S} \|M \cdot x_{\mathcal{C}_i}\|_2^2. \quad (5.4)$$

Proof. Fix S and put $z_i = \|y_i\|_2$ for $i \in S$. By Theorem 5.1,

$$\begin{aligned} \left| \left\| \sum_{i \in S} y_i \right\|_2^2 - \sum_{i \in S} \|y_i\|_2^2 \right| &\leq \sum_{\substack{i,j \in S \\ i \neq j}} |\langle y_i, y_j \rangle| \\ &\leq C \log(k/\epsilon) z^\top K_S z \\ &\leq C \log(k/\epsilon) \|K_S\|_{2 \rightarrow 2} \|z\|_2^2. \end{aligned}$$

The principal-submatrix form of Theorem 4.1 and (5.3) bound the last expression by $O(\epsilon) \sum_{i \in S} \|y_i\|_2^2$, proving (5.4). $\square$

6 Improved multiband approximation

Theorem 6.1 (Multiband approximation). *Assume an admissible localizer family. For every k -Fourier-sparse signal x and every $0 < \epsilon \leq 1/2$, there is a signal x' and intervals $I_1, \dots, I_n$, with $n \leq k$, such that*

$$\text{supp}(\hat{x}') \subseteq \bigcup_{i=1}^n I_i, \quad (6.1)$$

$$\sum_{i=1}^n |I_i| \leq C \frac{k^2 \log(k/\epsilon) (\log k)^{3/2}}{\epsilon}, \quad (6.2)$$

$$\|x - x'\|_{[-1,1]}^2 \leq \epsilon \|x\|_{[-1,1]}^2, \quad (6.3)$$

$$\|x'\|_{[-1,1]}^2 \geq (1 - \epsilon) \|x'\|_2^2. \quad (6.4)$$

Proof. By the standing constant-retuning convention, it suffices to establish the two energy guarantees with $O(\epsilon)$ error.

Run the clustering procedure with the value of d_{min} in (5.3), and define

$$\delta_i := \epsilon \frac{|\mathcal{C}_i|}{k}, \quad x' := \sum_{i=1}^n M_{|\mathcal{C}_i|, \delta_i} \cdot x_{\mathcal{C}_i}. \quad (6.5)$$

Approximation. Put

$$E_i := \|M \cdot x_{\mathcal{C}_i}\|_2^2.$$

By Item (L3), the triangle inequality, and Cauchy–Schwarz,

$$\begin{aligned} \|x' - M \cdot x\|_2 &\leq \sum_i \|(M_{|\mathcal{C}_i|, \delta_i} - M)x_{\mathcal{C}_i}\|_2 \\ &\leq \sqrt{A} \sum_i \sqrt{\delta_i E_i} \\ &\leq \sqrt{A \sum_i \delta_i} \sqrt{\sum_i E_i} = \sqrt{A\epsilon} \sqrt{\sum_i E_i}. \end{aligned}$$

Applying Theorem 5.2 with $S = [n]$ gives

$$\|x' - M \cdot x\|_2^2 \leq C\epsilon \|M \cdot x\|_2^2. \quad (6.6)$$

Property (L1) implies $\|M \cdot x\|_2^2 = O(\|x\|_{[-1,1]}^2)$. Combining this fact, (6.6), and the inside-window approximation in Item (L1) proves

$$\|x - x'\|_{[-1,1]}^2 \leq C\epsilon \|x\|_{[-1,1]}^2. \quad (6.7)$$

Localization. The outside-window part of Item (L1) and (6.6) give

$$\|x'\|_{\mathbb{R} \setminus [-1,1]}^2 \leq C\epsilon \|x\|_{[-1,1]}^2.$$

By (6.7), $\|x'\|_{[-1,1]} \geq \frac{1}{2} \|x\|_{[-1,1]}$ for sufficiently small ϵ . Hence

$$\|x'\|_{\mathbb{R} \setminus [-1,1]}^2 \leq C\epsilon \|x'\|_2^2. \quad (6.8)$$

Fourier support. By Item (L2), the i -th term of (6.5) is supported in

$$I_i := \text{range}(\mathcal{C}_i) \pm \Delta_{|\mathcal{C}_i|, \delta_i}.$$

Thus (6.1) holds and $n \leq k$. Also,

$$\begin{aligned} \sum_i \Delta_{|\mathcal{C}_i|, \delta_i} &\leq \sum_i A \frac{|\mathcal{C}_i|^2}{\delta_i} \log \frac{|\mathcal{C}_i|}{\delta_i} \\ &= \frac{Ak}{\epsilon} \log \frac{k}{\epsilon} \sum_i |\mathcal{C}_i| = O\left(\frac{k^2 \log(k/\epsilon)}{\epsilon}\right). \end{aligned} \quad (6.9)$$

On the other hand, Theorem 3.1 and (5.3) imply

$$\sum_i |\text{range}(\mathcal{C}_i)| = O\left(\frac{k^2 \log(k/\epsilon)(\log k)^{3/2}}{\epsilon}\right). \quad (6.10)$$

Equations (6.9) and (6.10) prove the width bound. The standing constant retuning turns (6.7) and (6.8) into (6.3) and (6.4). $\square$

Remark 6.2 (Sample complexity). *Any multiband learner whose sample complexity is linear, up to logarithms, in the total support width inherits*

$$\tilde{O}\left(\frac{k^2 \log(k/\epsilon)(\log k)^{3/2}}{\epsilon}\right)$$

samples from Theorem 6.1. This note does not reprove the multiband learning theorem itself.

7 Tightness of the envelope analysis

The $\sqrt{\log k}$ in Theorem 4.1 is not an artifact of the proof when the pairwise envelope is treated as an arbitrary nonnegative matrix.

Proposition 7.1 (A matching 2-adic envelope). *For arbitrarily large k , there are ordered cluster sizes with total mass at most k and locations satisfying $d_{i,j} \geq d_{\min} \min(s_i, s_j)$ such that the uncapped matrix $K_{ij} = s_i s_j / d_{i,j}$ obeys*

$$\|K\|_{2 \rightarrow 2} \geq c \frac{k \sqrt{\log k}}{d_{\min}}.$$

Proof. Let $Q_0 = \lfloor \log(k/(16 \log k)) \rfloor$ and $N = 2^{Q_0} - 1$. For $1 \leq m \leq N$, create a cluster of size

$$s_m := 2^{v_2(m)}$$

at location $(1 + o(1))d_{\min}m$, where $v_2(m)$ is the 2-adic valuation. For every $q < Q_0$, exactly 2^{Q_0-q-1} indices have valuation q , so the total side mass is $Q_0 2^{Q_0-1} \leq k/4$. Add a central cluster of size $a = \lfloor k/2 \rfloor$ at zero.

If $m \neq n$, then $|m - n|$ is a nonzero multiple of $2^{\min(v_2(m), v_2(n))}$. Thus all side-side separations are at least $d_{\min} \min(s_m, s_n)$. Also m is a positive multiple of s_m , so the central-side separation is at least $d_{\min} s_m$. An arbitrarily small perturbation makes all inequalities strict.

Writing $m = 2^q u$ with u odd gives

$$K_{0m} = \frac{a 2^q}{d_{\min} 2^q u} = \frac{a}{d_{\min} u}.$$

For every $q < Q_0$, the index $m = 2^q$ occurs and contributes $a/d_{\min}$. Hence the Euclidean norm of the central off-diagonal row is at least $ck\sqrt{Q_0}/d_{\min}$. The corresponding star submatrix has top eigenvalue equal to this row norm. Since K is entrywise nonnegative and dominates the star, monotonicity of the Perron eigenvalue proves the claim. $\square$

Remark 7.2. *Theorem 7.1 is a lower bound for the abstract entrywise envelope, not for the actual Fourier Gram matrix. A stronger many-cluster Fourier estimate could still exploit oscillation or positive-definiteness constraints that are absent from the pairwise certificate.*

8 Why product-size merging does not help

For comparison, consider the binary merge cost

$$T_\lambda(a, b) := \min\{\lambda ab, 2\Delta_k\}.$$

Proposition 8.1 (Product-merge span). *Without the cap, every binary merge tree on m singleton leaves has total product cost*

$$\lambda \binom{m}{2}.$$

With the cap, the worst-case cost is

$$\lambda \sum_{j=1}^{m-1} \min\left\{j, \frac{2\Delta_k}{\lambda}\right\} = \Theta(\min\{\lambda m^2, \Delta_k m\}).$$

Proof. In the uncapped case, the identity

$$\binom{a}{2} + \binom{b}{2} + ab = \binom{a+b}{2}$$

proves the claim by induction, independently of the tree. For the capped case, the comb tree that successively merges a singleton with the existing size- j cluster attains the displayed sum. The same sum is an upper bound by induction, using

$$\sum_{j=1}^{a+b-1} \min\{j, h\} - \sum_{j=1}^{a-1} \min\{j, h\} - \sum_{j=1}^{b-1} \min\{j, h\} \geq \min\{ab, h\}$$

with $h = 2\Delta_k/\lambda$. □

Proposition 8.2 (Energy obstruction for product termination). *Suppose a proof uses only the pairwise envelope*

$$C \log(k/\epsilon) \frac{|\mathcal{C}_i||\mathcal{C}_j|}{\text{dist}(\mathcal{C}_i, \mathcal{C}_j)}$$

and an absolute row-sum argument. A terminal condition $\text{dist}(\mathcal{C}_i, \mathcal{C}_j) > \lambda|\mathcal{C}_i||\mathcal{C}_j|$ cannot certify relative error $O(\epsilon)$ for all partitions unless

$$\lambda = \Omega\left(\frac{k \log(k/\epsilon)}{\epsilon}\right).$$

At that scale, Theorem 8.1 permits merge span $\Omega(k^3 \log(k/\epsilon)/\epsilon)$.

Proof. Take one cluster of size $m = \lfloor k/2 \rfloor$ at zero and m singleton clusters at positions

$$x_j = (1 + o(1))\lambda(m + 2j), \quad 1 \leq j \leq m.$$

This configuration satisfies every product-separation condition. The absolute envelope row of the large cluster has sum

$$C \log(k/\epsilon) \sum_{j=1}^m \frac{m}{x_j} = \Theta\left(\frac{k \log(k/\epsilon)}{\lambda}\right).$$

Requiring this to be $O(\epsilon)$ proves the lower bound on λ . Substituting that value into the uncapped product span $\lambda \binom{k}{2}$ proves the final claim. If, in addition, $2\Delta_k \geq c\lambda k$ for a sufficiently large absolute constant c , then the successive comb costs remain below the cap up to constants, so the capped construction has the same order. □

9 Discussion

The improvement in this note comes from using the geometry of the entire correlation matrix, not from changing which frequencies are merged. The dyadic block proof replaces a worst-row estimate by square summability within each size pair and then aggregates the size blocks in Frobenius norm.

The 2-adic example shows that this method has reached the limit of the current entrywise pairwise certificate. A further improvement must establish a many-cluster statement for the actual Fourier Gram matrix. Plausible inputs include an oscillatory cancellation estimate or a Bessel/frame inequality adapted to clusters of unequal sparsity. Neither follows from pairwise almost-orthogonality alone.